

Name: _____

Date: _____

SIMULTANEOUS EQUATIONS, (NON-LINEAR SIMULTANEOUS EQUATIONS)

LO: I can solve simultaneous equations where one equation is linear and one is quadratic.

Non-linear simultaneous equations

Non-linear simultaneous equations involve one linear equation and one equation with a **squared term** (e.g. circle or quadratic). Use substitution to solve.

Strategy: Rearrange the linear equation to make one variable the subject, substitute into the quadratic, then solve.

Quick examples

●	$y = x + 1$ and $x^2 + y^2 = 5$: substitute y	circle and line
●	$y = 2x$ and $y = x^2 - 3$: set equal	parabola and line
●	Usually gives two pairs of solutions	line crosses curve twice
●	Use elimination for non-linear equations	must use substitution

Key idea

Substitute the linear equation into the quadratic to get a single equation in one variable. Then solve the resulting quadratic.

$$y = x + 1; x^2 + \textcircled{y^2} = 5 \quad x^2 + \textcircled{(x+1)^2} = 5$$

Replace y with the linear expression, then solve

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - line and circle

Solve $y = x + 1$ and $x^2 + y^2 = 5$

$$\text{Substitute: } x^2 + (x+1)^2 = 5 \quad x^2 + x^2 + 2x + 1 = 5$$

$$2x^2 + 2x - 4 = 0 \quad x^2 + x - 2 = 0 \quad (x+2)(x-1) = 0$$

$$x = -2, y = -1 \text{ or } x = 1, y = 2$$

(-2, -1) and (1, 2)

Find x values from the quadratic, then substitute back for y .

Example 2 - line and parabola

Solve $y = 2x + 1$ and $y = x^2 - 2$

$$\text{Set equal: } x^2 - 2 = 2x + 1 \quad x^2 - 2x - 3 = 0 \quad (x-3)(x+1) = 0$$

$$x = 3, y = 7 \text{ or } x = -1, y = -1$$

(3, 7) and (-1, -1)

When both equations equal y , set the right-hand sides equal.

Example 3 - substitution approach

Solve $x + y = 5$ and $xy = 6$

$$\text{From linear: } y = 5 - x$$

$$\text{Substitute: } x(5-x) = 6 \quad 5x - x^2 = 6$$

$$x^2 - 5x + 6 = 0$$

$$(x-2)(x-3) = 0: x = 2, y = 3 \text{ or } x = 3, y = 2$$

(2, 3) and (3, 2)

Rearrange the linear equation for one variable, then substitute.

Example 4 - one solution

Solve $y = 2x - 1$ and $y = x^2 - 2x + 2$

$$\text{Set equal: } x^2 - 2x + 2 = 2x - 1 \quad x^2 - 4x + 3 = 0 \quad (x-1)(x-3) = 0$$

$$x = 1, y = 1 \text{ or } x = 3, y = 5$$

(1, 1) and (3, 5)

Check each solution satisfies both original equations.

YOUR TURN – PRACTICE (deliberate practice)

1. Line and parabola

Solve each pair simultaneously.

- a) $y = x + 2$ and $y = x^2$
- b) $y = 3x$ and $y = x^2 + 2$
- c) $y = x + 6$ and $y = x^2$
- d) $y = 2x + 3$ and $y = x^2 + x + 1$

2. Line and circle

Solve each pair simultaneously.

- a) $y = x$ and $x^2 + y^2 = 8$
- b) $y = x + 2$ and $x^2 + y^2 = 10$
- c) $x + y = 5$ and $x^2 + y^2 = 13$
- d) $y = 2x$ and $x^2 + y^2 = 20$

3. Product relationships

Solve these pairs.

- a) $x + y = 7$ and $xy = 12$
- b) $x + y = 10$ and $xy = 21$
- c) $x - y = 2$ and $xy = 15$
- d) $x + y = 9$ and $xy = 20$

4. Harder problems

Solve each pair simultaneously.

- a) $y = 2x - 1$ and $y = x^2 - 3x + 3$
- b) $x + 2y = 7$ and $x^2 + y^2 = 10$
- c) $y = x + 1$ and $x^2 + 2y^2 = 6$
- d) $2x + y = 5$ and $x^2 + y^2 = 5$

5. Mixed non-linear

Solve each system.

- a) $y = x - 1$ and $x^2 + y^2 = 25$
- b) $y = 3 - x$ and $y = x^2 - x - 1$
- c) $x + y = 4$ and $x^2 + 3y = 10$
- d) $y = x + 3$ and $2x^2 + y^2 = 27$
- e) $y = 2x$ and $x^2 - y = 3$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To solve non-linear simultaneous equations, use substitution ☐

Correct answer: _____

Reason: _____

b) You can use elimination for quadratic simultaneous equations ☐

Correct answer: _____

Reason: _____

c) A line and a circle can intersect at 0, 1, or 2 points ☐

Correct answer: _____

Reason: _____

d) Non-linear systems always have exactly 2 solutions ☐

Correct answer: _____

Reason: _____

e) After substitution, you solve the resulting quadratic equation ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The line $y = x + k$ is tangent to the circle $x^2 + y^2 = 8$. Find the possible values of k . (Hint: the discriminant = 0 for a tangent.)

Expression: _____

Simplified: _____

b) Solve $x - y = 2$ and $x^2 + xy + y^2 = 12$ simultaneously. Show all working clearly.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

